A Novel Dynamic Neural System for Nonconvex Portfolio Optimization With Cardinality Restrictions

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***Abstract*—The Markowitz model, a portfolio analysis model that won the Nobel Prize, lays the theoretical groundwork for modern finance. The transaction cost and the cardinality restriction, which were not covered in Markowitz model, are becoming increasingly important with the advent of high-frequency trading era. However, it remains a challenging problem to consider those constraints due to the nonconvex nature of the problem. A novel dynamic neural network, inspired by its successes in machine learning, is developed to tackle this difficult issue. Theoretical analysis is provided for the convergence of the designed neural network. Experimental results using real stock market data confirm the effectiveness of the proposed model. With the proposed model, the cost function characterizing the overall risks, and rewards is reduced by 123.6% from -4.549×10^{-5} to -1.0173×10^{-4}. This indicates that the proposed strategy is successful in reducing risks and increasing rewards.**

***Index Terms*—Cardinality constraint, dynamic neural network, Markowitz model, nonconvex optimization, portfolio optimization.**

I. Introduction

STOCK market investment is one of the most common types of investments in our daily lives. The choice of which stocks to short or to long for the greatest profit is one of the main issues. From a financial standpoint, it is a game of risk versus reward. It is a central problem to optimally distribute funds among stocks in a way that minimizes risk and maximizes profit. In his ground-breaking work, Markowitz treated stock prices as a random variable and characterize the risk and reward statistically by variance and mean values, and formulate the portfolio weight determination as a quadratic optimization problem with both linear equality and linear inequality constraints. This formal framework offers a useful yet manageably model for practitioners to make logical financial judgments. Due to his seminal work, Markowitz received the 1990 Nobel prize in economics [1] and the development of this model constitutes the basis for modern portfolio theory [2], [3].

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In Markowitz's seminal work, he simplified the portfolio optimization problem by disregarding certain factors and formulating it as a convex optimization problem [1]. However, in the current era of high-frequency trading, transaction costs play a crucial role in determining the feasibility of an investment strategy. Despite this, Markowitz's model does not take transaction costs into consideration. Moreover, institutional investors often face the challenge of selecting a specific set of stocks to invest in, subject to a cardinality constraint. However, this constraint is inherently nonconvex and hence not accounted for in Markowitz's model. With the growing importance of transaction costs and cardinality constraints in portfolio optimization, there is a need for new computational tools to solve these problems. Recently, neural networks have gained significant attention as a powerful tool for financial data analysis and signal processing. Inspired by the success of neural networks, this work aims to extend their application to dynamic portfolio optimization problems with transaction costs and cardinality constraints, specifically in the area of high-frequency trading finance. This will address the current gaps in traditional portfolio optimization methods and provide a more accurate representation of real-world investment scenarios. Numerous studies have highlighted the potential of neural networks in finance [4], [5], [6], [7], [8], [9], and it is a consensus to push forward conventional finance by leveraging emerging computational methods [10], [11], [12], [13], [14], [15]. This work aims to leverage these advancements to push the boundaries of portfolio optimization.

The development of alternative models for performance improvement has been an important research area since the invention of the Markowitz model for portfolio analysis. Several limitations of conventional portfolio optimizer, e.g., the absence of the transaction cost, are identified in [16], [17], and [18] and a novel model is proposed with more realistic factors considered. Methods developed in [19] and [20] directly put the cardinality constraint into consideration in their formulated problem. However, it happens very often for the solution to get stuck at the local optima due to the nonconvex nature of the problem. Another solution to deal with the cardinality constraint is the relaxation of it to L_1 constraint, which degrades to convex and can be solved effectively using recent works on sparsity optimization [21]. However, this relaxation from nonconvex to convex inevitably alters the structure of the original problem and is usually at the scarification of the optimality of the solution. Evolutionary computation is widely used to deal with nonconvex optimization. Some researchers explore the possibility of using evolutionary computation, including particle swarm optimization [22], [23], [24], genetic algorithm [25], [26], and beetle antennae search [27], [28], [29], [30], to find the optimal portfolio under cardinality constraints. Although this type of strategy mostly is able to find the global optimum solution, it is usually at a high cost of computational burden and might not be suitable for high-frequency trading for which the processing time is critical.

Dynamic neural networks is a type of recurrent model with an internal feedback mechanism and is therefore suitable for real-time processing. The internal feedback mechanism of recurrent neural networks endow it with the ability to correct its error in the present state and escape from local optima. A recurrent neural network with the ability to invert a matrix with time-varying elements is proposed in [31] and [32], and later extended to its discrete-time counterparts [33], [34], [35], [36], [37]. Note that this type of model is able to solve a general class of nonconstrained optimization problems, including linear equation set and simplified portfolio optimization without cardinality constraint can also be solved using these methods. Novel designs of the activation functions are presented in [38], [39], and [40] to accelerate the convergence of recurrent neural networks. Complex-valued dynamic neural networks are investigated in [41], which includes the real-valued neural dynamics as a special case. In face of real world problems, noises and disturbances are usually inevitable and noise-robust neural network are thus developed in [42] and [43]. Other development of dynamic neural networks includes the establishment of data driven model-free model [44], the dedicated solver for Sylvester equation problem [45], nonstationary quadratic programs [46] and rank-deficit problem solving [47], the applications to robot arm motion control [48], electromagnetic suspension systems [49], mobile robot trajectory planning [50], electromechanical systems [51], and multiple robot coordination [52], [53].

Motivated by the success of dynamic neural networks in real-time processing, its ability to escape from local optima and the demand for portfolio optimization with both transaction cost and cardinality constraint in the high-frequency trading era, we present a novel dynamic neural model in this work to address the portfolio optimization problem. This neural network is proved rigorously for its convergence. The main contributions of this article are summarized as follows.

1) Compared to evolutionary optimization-based solutions, the proposed method does not require high computational power and therefore is suitable for high-frequency trading.
2) The optimality and global convergence of the proposed model are guaranteed in theory.
3) To the best of our knowledge, this is the first dynamic neural model developed for portfolio analysis with the consideration of the nonconvex cardinality constraint.
4) As verified on DJI stock data, the proposed method reduces the overall cost by 123.6% in comparison to the average performance of all the stock combinations with requested cardinality with optimal portfolio weights.

The remainder of this article is organized as follows. In Section II, some prior knowledge and mathematical tools used throughout this article are presented. In Section III, the portfolio optimization problem is described. The neural network is then designed to solve the formulated problem in Section IV and theoretical analysis is illustrated in Section V. Experimental validations based on real stock data are presented in Section VI. Finally, conclusions are drawn in Section VII.

II. Prior Knowledge

A sigmoid function $y = g(x)$ is defined as

$$g(x) = \frac{1}{1+e^{-cx}} \tag{1}$$

for $x \in \mathbb{R}$ where $c > 0$ is a constant. As an exponential function, e^{-cx} ranges in $(0, \infty)$, with which we can conclude the range of the sigmoid function $g(x)$ as

$$0 < g(x) < 1, \text{for any } x. \tag{2}$$

The derivative of $g(\cdot)$ can be calculated as

$$\frac{\partial g}{\partial x} = \frac{c}{1+e^{-cx}}\frac{e^{-cx}}{1+e^{-cx}} = cg(x)(1-g(x)) > 0 \tag{3}$$

which implies that the sigmoid function (1) is monotonically increasing.

For a vector $x \in \mathbb{R}^n$, its diagonalization matrix $A = \text{diag}(x)$ for $A \in \mathbb{R}^{n\times n}$ is defined as

$$A_{ij} = \begin{cases} x_i, & \text{for } i = j \\ 0, & \text{for } i \neq j. \end{cases} \tag{4}$$

For a entrywise mapping $y = f(x)$ from $x = [x_i] \in \mathbb{R}^n$ to $y = [y_i] \in \mathbb{R}^n$ with $y_i = f(x_i)$, where x_i and y_i are the ith element of vector x and y, respectively, its gradient matrix

$$\frac{\partial y}{\partial x} = \text{diag}\begin{bmatrix} \frac{\partial y_1}{\partial x_1} \\ \frac{\partial y_2}{\partial x_2} \\ \cdots \\ \frac{\partial y_n}{\partial x_n} \end{bmatrix} = \text{diag}\begin{bmatrix} \frac{\partial f}{\partial x_1} \\ \frac{\partial f}{\partial x_2} \\ \cdots \\ \frac{\partial f}{\partial x_n} \end{bmatrix}. \tag{5}$$

For two vectors $x = [x_i] \in \mathbb{R}^n$ and $y = [y_i] \in \mathbb{R}^n$, their elementwise multiplication $z = x \circ y$ is defined as $z_i = x_i y_i$ for $z = [z_i] \in \mathbb{R}^n$. The elementwise multiplication is commutative in the sense that $x \circ y = y \circ x$. The elementwise multiplication can be expressed in the matrix multiplication form with the aid of the diagonalization matrix in the form $x \circ y = \text{diag}(x)y = \text{diag}(y)x$.

III. Problem Definition and Reformulation

Consider the purchase of a collection of n stocks with x_i being the proportion invested to the ith stock with $0 \leq x_i \leq 1$. The portfolio optimization problem can be expressed as follows by combining all x_i together to generate a vector $x = [x_i] \in \mathbb{R}^n$:

$$\begin{aligned} \min_{x,z}\ & x^{\mathrm{T}}\Sigma x, \max_{x,z} x^{\mathrm{T}} r_0 \\ \text{s.t.}\ & \mathbf{1}^{\mathrm{T}} x + {t_0}^{\mathrm{T}} x = 1 \\ & \mathbf{1}^{\mathrm{T}} z = k_0 \\ & 0 \leq x \leq z \\ & z \in \{0, 1\}^n \end{aligned} \tag{6}$$

where $\Sigma \in \mathbb{R}^{n\times n}$ is the variance matrix of the stocks, $r_0 \in \mathbb{R}^n$ is the mean return of all stocks, $t_0 \in \mathbb{R}^n$ is the transaction cost rate, $\mathbf{1} \in \mathbb{R}^n$ is a vector with all elements being 1, $z \in \mathbb{R}^n$ is a boolean vector. The value of z can be either 0 or 1 to force the cardinality condition, i.e., the selection of at most k_0 stocks for investment. The goal of the optimization problem is to maximise profit while minimizing risk, which is represented by the portfolio's mean return and measured as variance, respectively. In actuality, weighted sum is typically used to balance risk and return, resulting in the following objective function being optimized:

$$\begin{aligned}\min_{x,z}\ & J(x,z) = x^{\mathrm{T}}\Sigma x - c_0 x^{\mathrm{T}} r_0\\ \text{s.t.}\ & \mathbf{1}^{\mathrm{T}}x + {t_0}^{\mathrm{T}}x = 1\\ & \mathbf{1}^{\mathrm{T}}z = k_0\\ & 0 \le x \le z\\ & z \in \{0,1\}^n \end{aligned}\tag{7}$$

where $c_0 > 0$ is a tradeoff coefficient. Optimization (7) degrades into the standard Markowitz portfolio optimization by removing the cardinality restriction. The cardinality constraints, although they make more sense from a financial standpoint by modeling the restrictions for the number of picked stocks, greatly raise the solution difficulty since they turn the normal Markowitz model from a convex optimization into a nonconvex problem. Instead of looking at this nonconvex problem directly, we shift to look at the problem's relaxed reformulation for useful answers.

Remark 1: The expression (6) describes the desire of investors to maximize the profit and minimize the risk. However, the objectives in (6) usually cannot be achieved simultaneously. The expression in (7) is a tractable reformulation of (6) and is implementable from optimization perspective.

To make the solution of the problem tractable, the constraint $z = [z_i] \in \{0,1\}^n$ is converted into an equivalent form as $z_i(z_i-1) = 0$. Remarkably that $z_i(z_i-1)$ is quadratic in z_i and becomes maximum when $z_i = 1$ or $z_i = 0$ for z_i restricted in the range $z_i \in [0,1]$. This inspires us to add $-c_1\sum_{i=1}^{n} z_i(z_i-1) = -c_1 z^{\mathrm{T}}z + c_1\mathbf{1}^{\mathrm{T}}z$ with $c_1 > 0$ to the objective function to be minimized for the enforcement of the constraint $z \in \{0,1\}^n$. A reformulated problem is thus stated as

$$\begin{aligned}\min_{x,z}\ & J(x,z) = x^{\mathrm{T}}\Sigma x - c_0 x^{\mathrm{T}} r_0 - c_1 z^{\mathrm{T}}z + c_1\mathbf{1}^{\mathrm{T}}z\\ \text{s.t.}\ & \mathbf{1}^{\mathrm{T}}x + {t_0}^{\mathrm{T}}x = 1\\ & \mathbf{1}^{\mathrm{T}}z = k_0\\ & 0 \le x \le z\\ & 0 \le z \le 1. \end{aligned}\tag{8}$$

Equality constraints $\mathbf{1}^{\mathrm{T}}x + {t_0}^{\mathrm{T}}x = 1$ and $\mathbf{1}^{\mathrm{T}}z = k_0$ still show up in the optimization. Penalty terms $(\mathbf{1}^{\mathrm{T}}x + {t_0}^{\mathrm{T}}x - 1)^2$ and $(\mathbf{1}^{\mathrm{T}}z - k_0)^2$ are included in the objective function to relax the requirement of the explicit equality constraints. We define the following sigmoid function by introducing the auxiliary variable $y_1 \in \mathbb{R}^n$ to equalize the inequality constraint $0 \le z \le 1$:

$$z = g(y_1)\tag{9}$$

where the sigmoid function $g(\cdot)$ is defined in (1) with coefficient $c = c_3 > 0$. Due to this definition, z has all its values ranging between 0 and 1. As z is expressed in the form of $g(y_1)$, i.e., as a sigmoid function, we can reach the strict satisfaction of the inequality constraint $0 \le z \le 1$. Similarly, the inequality $0 \le x \le z$ can be equivalently written as $x = z \circ g(y_2)$ with $g(\cdot)$ defined in (1) and "$\circ$" represents the elementwise multiplication with $y_2 \in \mathbb{R}^n$ being an auxiliary variable. With the new expressions of z and x in terms of sigmoid functions $z = g(y_1)$ and $x = z \circ g(y_2)$, we can reformulate (8) into the following form without the appearance of any explicit constraints:

$$\begin{aligned}\min_{y_1,y_2}\ & J(y_1,y_2) = x^{\mathrm{T}}\Sigma x - c_0 x^{\mathrm{T}} r_0 - c_1 z^{\mathrm{T}}z + c_1\mathbf{1}^{\mathrm{T}}z\\ & + c_2\big(\mathbf{1}^{\mathrm{T}}x + {t_0}^{\mathrm{T}}x - 1\big)^2 + c_2\big(\mathbf{1}^{\mathrm{T}}z - k_0\big)^2\\ & \text{with}\ z = g(y_1),\ x = z \circ g(y_2) \end{aligned}\tag{10}$$

where $y_1 \in \mathbb{R}^n$ and $y_2 \in \mathbb{R}^n$ are the two auxiliary variables associated with the constraints on z and x, respectively. So far we have reformulated approximately the portfolio optimization problem with the consideration of cardinality constraints and transaction costs into a nonlinear optimization without explicit constraints.

About the conversion of the constraint $0 \le x \le z$ in (8) to $z = g(y_1)$ in (10), we have the following remark.

Remark 2: According to (2), $z = g(y_1)$ leads to the range of z as $0 < z < 1$ in contrast to the original constraint $0 \le z \le 1$ in (8). Note that the value of $z = g(y_1)$ cannot achieve 0 or 1 but it can be indefinitely close to 0 or 1 when y_1 goes to $-\infty$ or ∞, implying the fact that the error can be indefinitely small for converting $0 \le z \le 1$ in (8) to $z = g(y_1)$ in (10).

IV. Neural Network Design

In this section, a novel neural network is designed to solve the portfolio optimization problem formulated in (10), i.e., the minimization of the objective function $J(y_1, y_2)$ relative to the decision variables y_1 and y_2.

We first derive the expression of the derivatives of the objective function in (10). The derivative of the objective function in (10) relative to y_1 can be expressed as

$$\begin{aligned}\frac{\partial J}{\partial y_1} &= 2\frac{\partial^{\mathrm{T}}x}{\partial y_1}\Sigma x - c_0\frac{\partial^{\mathrm{T}}x}{\partial y_1}r_0 - 2c_1\frac{\partial^{\mathrm{T}}z}{\partial y_1}z + c_1\frac{\partial^{\mathrm{T}}z}{\partial y_1}\mathbf{1}\\ &\quad + 2c_2\big(\mathbf{1}^{\mathrm{T}}x + {t_0}^{\mathrm{T}}x - 1\big)\left(\frac{\partial^{\mathrm{T}}x}{\partial y_1}\mathbf{1} + \frac{\partial^{\mathrm{T}}x}{\partial y_1}t_0\right)\\ &\quad + 2c_2\big(\mathbf{1}^{\mathrm{T}}z - k_0\big)\frac{\partial^{\mathrm{T}}z}{\partial y_1}\mathbf{1}\\ &= \frac{\partial^{\mathrm{T}}x}{\partial y_1}\big(2\Sigma x - c_0 r_0 + 2c_2\big(\mathbf{1}^{\mathrm{T}}x + {t_0}^{\mathrm{T}}x - 1\big)(\mathbf{1} + t_0)\big)\\ &\quad + \frac{\partial^{\mathrm{T}}z}{\partial y_1}\big(-2c_1 z + c_1\mathbf{1}\big). \end{aligned}\tag{11}$$

The above equations include $(\partial^{\mathrm{T}}x/\partial y_1)$ and $(\partial^{\mathrm{T}}z/\partial y_1)$, which can be further expanded based on the definitions of x and z in (10)

$$\begin{aligned}\frac{\partial z}{\partial y_1} &= \frac{\partial g}{\partial y_1}\\ \frac{\partial x}{\partial y_1} &= \frac{\partial(z\circ g(y_2))}{\partial y_1} = \frac{\partial(z\circ g(y_2))}{\partial y_1}\\ &= \operatorname{diag}(g(y_2))\frac{\partial z}{\partial y_1} = \operatorname{diag}(g(y_2))\frac{\partial g}{\partial y_1}. \end{aligned}\tag{12}$$

The substitution of (12) into (11) yields

$$\begin{aligned}\frac{\partial J}{\partial y_1} &= \frac{\partial^{\mathrm{T}} g}{\partial y_1}\big(\operatorname{diag}(g(y_2))\big(2\Sigma x - c_0 r_0\\ &\quad + 2c_2\big(\mathbf{1}^{\mathrm{T}}x + {t_0}^{\mathrm{T}}x - 1\big)(\mathbf{1}+t_0)\big)\\ &\quad - 2c_1 z + c_1\mathbf{1} + 2c_2\big(\mathbf{1}^{\mathrm{T}}z - k_0\big)\mathbf{1}\big)\\ &= \frac{\partial^{\mathrm{T}} g}{\partial y_1}\big(g(y_2)\circ\big(2\Sigma x - c_0 r_0\\ &\quad + 2c_2\big(\mathbf{1}^{\mathrm{T}}x + {t_0}^{\mathrm{T}}x - 1\big)(\mathbf{1}+t_0)\big)\\ &\quad - 2c_1 z + c_1\mathbf{1} + 2c_2\big(\mathbf{1}^{\mathrm{T}}z - k_0\big)\mathbf{1}\big)\\ &= \psi(y_1)\circ\big(g(y_2)\circ\big(2\Sigma x - c_0 r_0\\ &\quad + 2c_2\big(\mathbf{1}^{\mathrm{T}}x + {t_0}^{\mathrm{T}}x - 1\big)(\mathbf{1}+t_0)\big)\\ &\quad - 2c_1 z + c_1\mathbf{1} + 2c_2\big(\mathbf{1}^{\mathrm{T}}z - k_0\big)\mathbf{1}\big)\end{aligned}\tag{13}$$

where $\psi(x_i)$ is the ith element of $\psi(x)$ for $x = [x_i]$ can be calculated as

$$\psi(x_i) = \frac{\partial g(x_i)}{\partial x_i} = \frac{c_4 e^{-c_4 x_i}}{\left(1+e^{-c_4 x_i}\right)^2} = c_4 g(x_i)(1-g(x_i)).$$

Similar treatment applied to the derivative relative to y_2 with the notice of $\partial z/\partial y_2 = \partial g(y_1)/\partial y_2 = 0$ results in

$$\begin{aligned}\frac{\partial J}{\partial y_2} &= \frac{\partial^{\mathrm{T}} x}{\partial y_2}\big(2\Sigma x - c_0 r_0 + 2c_2\big(\mathbf{1}^{\mathrm{T}}x + {t_0}^{\mathrm{T}}x - 1\big)(\mathbf{1}+t_0)\big)\\ &= \frac{\partial^{\mathrm{T}}(z\circ g(y_2))}{\partial y_2}\big(2\Sigma x - c_0 r_0 + 2c_2\big(\mathbf{1}^{\mathrm{T}}x + {t_0}^{\mathrm{T}}x - 1\big)\\ &\quad\cdot(\mathbf{1}+t_0)\big)\\ &= \frac{\partial^{\mathrm{T}} g}{\partial y_2}\operatorname{diag}(z)\big(2\Sigma x - c_0 r_0 + 2c_2\big(\mathbf{1}^{\mathrm{T}}x + {t_0}^{\mathrm{T}}x - 1\big)\\ &\quad\cdot(\mathbf{1}+t_0)\big)\\ &= \frac{\partial^{\mathrm{T}} g}{\partial y_2}\big(z\circ\big(2\Sigma x - c_0 r_0 + 2c_2\big(\mathbf{1}^{\mathrm{T}}x + {t_0}^{\mathrm{T}}x - 1\big)\\ &\quad\cdot(\mathbf{1}+t_0)\big)\big)\\ &= \psi(y_2)\circ\big(z\circ\big(2\Sigma x - c_0 r_0 + 2c_2\big(\mathbf{1}^{\mathrm{T}}x + {t_0}^{\mathrm{T}}x - 1\big)\\ &\quad\cdot(\mathbf{1}+t_0)\big)\big).\end{aligned}\tag{14}$$

An intuitive design of a dynamic system to minimize J is to define negative gradient dynamics as below

$$\begin{aligned}\epsilon\dot{y}_1 &= -\frac{\partial J}{\partial y_1} = -\psi(y_1)\circ\big(g(y_2)\circ\big(2\Sigma x - c_0 r_0\\ &\quad + 2c_2\big(\mathbf{1}^{\mathrm{T}}x + {t_0}^{\mathrm{T}}x - 1\big)(\mathbf{1}+t_0)\big)\\ &\quad - 2c_1 z + c_1\mathbf{1} + 2c_2\big(\mathbf{1}^{\mathrm{T}}z - k_0\big)\mathbf{1}\big)\\ \epsilon\dot{y}_2 &= -\frac{\partial J}{\partial y_2} = -\psi(y_2)\circ\big(z\circ\big(2\Sigma x - c_0 r_0\\ &\quad + 2c_2\big(\mathbf{1}^{\mathrm{T}}x + {t_0}^{\mathrm{T}}x - 1\big)(\mathbf{1}+t_0)\big)\big).\end{aligned}$$

Note that $\psi(y_1)$ and $\psi(y_2)$ appear in the above negative gradient dynamics. Instead, the following model without the appearance of $\psi(y_1)$ and $\psi(y_2)$ in its dynamics can be designed for the solution of portfolio optimization by noticing the non-negativeness of $\psi(\cdot)$:

State Equation

$$\begin{cases}\epsilon\dot{y}_1 = -g(y_2)\circ\big(2\Sigma x - c_0 r_0\\ \qquad + 2c_2\big(\mathbf{1}^{\mathrm{T}}x + {t_0}^{\mathrm{T}}x - 1\big)(\mathbf{1}+t_0)\big) + 2c_1 z - c_1\mathbf{1}\\ \qquad - 2c_2\big(\mathbf{1}^{\mathrm{T}}z - k_0\big)\mathbf{1}\\ \epsilon\dot{y}_2 = -\big(2\Sigma x - c_0 r_0 + 2c_2\big(\mathbf{1}^{\mathrm{T}}x + {t_0}^{\mathrm{T}}x - 1\big)\\ \qquad\cdot(\mathbf{1}+t_0)\big).\end{cases}$$

Output Equation

$$\begin{cases} z = g(y_1)\\ x = z\circ g(y_2).\end{cases}$$

Notice that the term $2\Sigma x - c_0 r_0 + 2c_2(\mathbf{1}^{\mathrm{T}}x + {t_0}^{\mathrm{T}}x - 1)(\mathbf{1}+t_0)$ appears in the dynamics of both y_1 and y_2 which allows us to reach a simplified form for the dynamics of y_1 as

$$\begin{aligned}\epsilon\dot{y}_1 &= \epsilon g(y_2)\circ\dot{y}_2 + 2c_1 z - c_1\mathbf{1} - 2c_2\big(\mathbf{1}^{\mathrm{T}}z - k_0\big)\mathbf{1}\\ &= \epsilon\dot{\phi}(y_2) + 2c_1 z - c_1\mathbf{1} - 2c_2\big(\mathbf{1}^{\mathrm{T}}z - k_0\big)\mathbf{1}\end{aligned}\tag{15}$$

where $\phi(\cdot)$ is defined as follows

$$\phi(x) = \begin{bmatrix}\phi(x_1)\\ \phi(x_2)\\ \vdots\\ \phi(x_n)\end{bmatrix} \text{for } x = \begin{bmatrix}x_1\\ x_2\\ \vdots\\ x_n\end{bmatrix}\in\mathbb{R}^n$$

with

$$\phi(x_i) = \int_0^{x_i} g(x_i)dx_i = \frac{\ln(1+e^{c_3 x_i})}{c_3}.\tag{16}$$

Based on the above definition of $\phi(\cdot)$, it can be easily verified that $\dot{\phi}(y_2) = g(y_2)\circ\dot{y}_2$ as utilized in (15). By defining an extra auxiliary variable $y_3\in\mathbb{R}^n$ as $y_3 = y_1 - \phi(y_2)$, (15) can be rewritten as

$$\epsilon\dot{y}_3 = 2c_1 z - c_1\mathbf{1} - 2c_2\big(\mathbf{1}^{\mathrm{T}}z - k_0\big)\mathbf{1}\tag{17}$$

with the definition of y_1 as

$$y_1 = \phi(y_2) + y_3.\tag{18}$$

We thus are ready to present a simplified model in its complete form as

Dynamic Neural Network for Portfolio Optimization: (19)

State Equation

$$\begin{cases}\epsilon\dot{y}_2 = -2\Sigma x + c_0 r_0 - 2c_2\big(\mathbf{1}^{\mathrm{T}}x + {t_0}^{\mathrm{T}}x - 1\big)\\ \qquad\cdot(\mathbf{1}+t_0),\\ \epsilon\dot{y}_3 = 2c_1 z - c_1\mathbf{1} - 2c_2\big(\mathbf{1}^{\mathrm{T}}z - k_0\big)\mathbf{1}.\end{cases}$$

Output Equation

$$\begin{cases} y_1 = \phi(y_2) + y_3\\ z \ = g(y_1)\\ x \ = z\circ g(y_2).\end{cases}$$

For the above derived dynamic neural network, the state variables y_2 and y_3 evolve dynamically with the feedback from the output x, y_1 and z of this network. The schematics of the model is shown in Fig. 1, from which it is observable that the feedback information from the outputs x and z, together with the inputs r_0, k_0, t_0, and Σ to determine the value of the

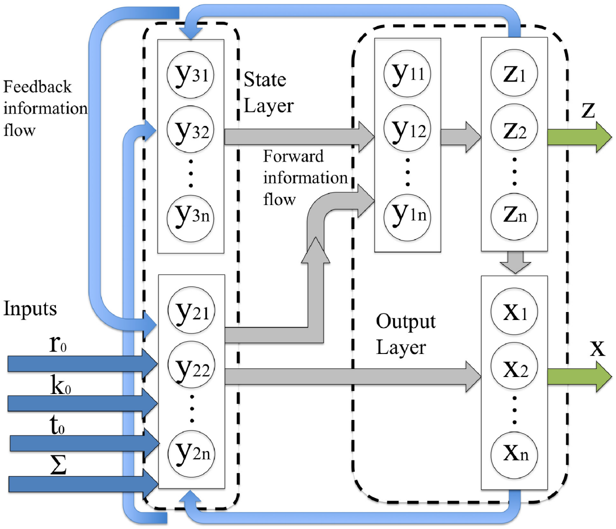

Fig. 1. Schematics of the proposed dynamic neural network for cardinality constrained portfolio optimization with the consideration of transaction costs.

state variables y_2 and y_3. The dynamic dependence of the state variables on both the input and the output evidently implies that the proposed model is indeed a dynamic neural network. The proposed model is used to solve x and z recurrently, which are the optimal solution to the cardinality constrained portfolio optimization (10).

V. Theoretical Analysis

In this section, we present theoretical analysis to the neural model expressed in (19).

About the compliance of the output obtained from the proposed dynamic neural network (19) to the inequality constraints in (10), we have the following lemma.

Lemma 1: The output variables x and z in the dynamic neural network (19) satisfy the inequalities in (10).

Proof: As $z = g(y_1)$ and it is evident that the range of function $g(\cdot)$ is $[0, 1]$, we conclude that $0 \le z \le 1$. For $x = z \circ g(y_2)$, as the range of $g(y_2)$ is $[0, 1]$, it is obtained that $0 \le x \le z$, which implies that x and z obtained in (19) satisfy the inequality constraints in (10). This completes the proof. ■

About the optimality of the solution obtained from model (19) to the optimization (10), we have the following lemma.

Lemma 2: The optimal solution of (10) is an equilibrium point of the proposed dynamic neural network (19).

Proof: First, we can solve the equilibrium point of model (19) by letting the right-hand side of its state equation, denoted as η_1 and η_2 to be 0

$$\begin{aligned} 0 &= \eta_1 = -2\Sigma x + c_0 r_0 - 2c_2\big(\mathbf{1}^{\mathrm{T}} x + {t_0}^{\mathrm{T}} x - 1\big) \cdot (\mathbf{1} + t_0), \\ 0 &= \eta_2 = 2c_1 z - c_1 \mathbf{1} - 2c_2\big(\mathbf{1}^{\mathrm{T}} z - k_0\big)\mathbf{1}. \end{aligned} \tag{20}$$

To prove that this solution includes the optimal solution of (10), we only need to show that the solution of (20) is identical to the following gradient conditions for optimality of (10):

$$\frac{\partial J}{\partial x} = 0,\ \frac{\partial J}{\partial z} = 0,\ 0 \le x \le z,\ 0 \le z \le 1. \tag{21}$$

In Lemma 1, it has been proved that the equilibrium point of (19) satisfies the inequality constraints above, and therefore we only need to show the gradient condition in (21) holds. The detailed expression of $\partial J/\partial x$ and $\partial J/\partial z$ can be obtained from the expression of $J(x, z)$ in (10) as

$$\begin{aligned} \frac{\partial J}{\partial x} &= 2\Sigma x - c_0 r_0 + 2c_2\big(\mathbf{1}^{\mathrm{T}} x + {t_0}^{\mathrm{T}} x - 1\big)(\mathbf{1} + t_0) \\ \frac{\partial J}{\partial z} &= -2c_1 z + c_1 \mathbf{1} + 2c_2\big(\mathbf{1}^{\mathrm{T}} z - k_0\big)\mathbf{1}. \end{aligned} \tag{22}$$

Comparing (20) with (22), it is immediately reached that

$$\frac{\partial J}{\partial x} = -\eta_1 = 0,\ \frac{\partial J}{\partial z} = -\eta_2 = 0 \tag{23}$$

which concludes that the optimal solution of (10) is an equilibrium point of model (19). ■

In Lemma 2, it has been stated that the optimal solution of (10) is an equilibrium point of the neural model. The model developed in this article is essentially an ordinary differential equation and it is necessary to prove the convergence of the neural model to guarantee that it evolves toward its equilibrium point with time whatever initial values are taken. The following lemma gives a statement on this global convergence property of dynamic neural network (19).

Lemma 3: Variables x and z in the proposed dynamic neural network (19) globally converge to the values corresponding to one equilibrium point of it.

Proof: The substitution of $y_3 = y_1 - \phi(y_2)$ into $\dot{y}_3$ turns (19) into (15), which implies the global convergence can be achieved by proving that of (15). Define a Lyapunov function as $V = J(y_1, y_2) = x^{\mathrm{T}}\Sigma x - c_0 x^{\mathrm{T}} r_0 - c_1 z^{\mathrm{T}} z + c_1 \mathbf{1}^{\mathrm{T}} z + c_2(\mathbf{1}^{\mathrm{T}} x + {t_0}^{\mathrm{T}} x - 1)^2 + 2c_2(\mathbf{1}^{\mathrm{T}} z - k_0)^2$. Since we have already proved in Lemma 1 that x and z always stay in the constrained set $0 \le z \le 1$, $0 \le x \le z \le 1$, the Lyapunov V, as expressed as a continuous function of x and z with x and z in a bounded region defined by $x \in [0, 1]$ and $z \in [0, 1]$, is thus bounded from below. The time derivative of V along the dynamics of y_1 and y_2 can be calculated as

$$\dot{V} = \frac{\partial^{\mathrm{T}} J}{\partial y_1}\dot{y}_1 + \frac{\partial^{\mathrm{T}} J}{\partial y_2}\dot{y}_2. \tag{24}$$

Substituting the expressions of $(\partial J/\partial y_1)$ in (13) and $(\partial J/\partial y_2)$ in (14) to (24) and the expressions of $\dot{y}_1$ and $\dot{y}_2$ yields

$$\begin{aligned} \dot{V} &= -\alpha_1^{\mathrm{T}} \mathrm{diag}(\psi(y_1))\alpha_1 - \alpha_2^{\mathrm{T}} \mathrm{diag}(\psi(y_2) \circ z)\alpha_2 \\ &= -\alpha_1^{\mathrm{T}} \mathrm{diag}(\psi(y_1))\alpha_1 - \alpha_2^{\mathrm{T}} \mathrm{diag}(\psi(y_2) \circ g(y_1))\alpha_2 \\ &\le 0 \end{aligned} \tag{25}$$

where $\alpha_1 = g(y_2) \circ (2\Sigma x - c_0 r_0 + 2c_2(\mathbf{1}^{\mathrm{T}} x + {t_0}^{\mathrm{T}} x - 1)(\mathbf{1} + t_0)) - 2c_1 z + c_1 \mathbf{1} + 2c_2(\mathbf{1}^{\mathrm{T}} z - k_0)\mathbf{1}$ and $\alpha_2 = (2\Sigma x - c_0 r_0 + 2c_2(\mathbf{1}^{\mathrm{T}} x + {t_0}^{\mathrm{T}} x - 1)(\mathbf{1} + t_0))$. Note that both $\mathrm{diag}(\psi(y_1))$ and $\mathrm{diag}(\psi(y_2) \circ g(y_1))$ are positive semi-definite since $\psi(x) > 0$ and $g(x) > 0$ for any finite value of x and therefore $\dot{V} \le 0$ is achieved in (25). Following LaSalle's principle, we conclude that the system converges globally to the largest invariant set formed by:

$$\begin{aligned} \dot{V} = 0 &\iff \psi(y_1) \circ \alpha_1 = 0,\ \psi(y_2) \circ g(y_1) \circ \alpha_2 = 0 \\ &\iff \alpha_1 = 0, \alpha_2 = 0. \end{aligned} \tag{26}$$

Remarkably that $\alpha_1 = 0$ and $\alpha_2 = 0$ are exactly the equilibrium point of (15), it is thus concluded for the global convergence of (15) to its equilibrium point. The inherent equivalent of (15) and (19) therefore implies the global convergence of (19) to its equilibrium point, which concludes the proof of this lemma. ■

Combining the conclusions in Lemmas 1–3, the following main theorem can be obtained.

Theorem 1: Variables x and z in the proposed dynamic neural network (19) globally converge to the values corresponding to the equilibrium point which includes the global optimum of the portfolio optimization under the cardinality and the transaction cost constraints (10).

Proof: This is an immediate result combining conclusions in Lemmas 1–3. ■

Remark 3: The portfolio optimization (10) is nonconvex due to the presence of the cardinality constraint and local optima may exists. The neural model developed in this article evolves toward the saddle point set of (10) which includes the global optima in it.

VI. Numerical Experiments

In this section, we present the experimental results based on real data for portfolio optimization to show the effectiveness of the proposed method in this article.

A. Illustrative Experiments With Small Number of Stocks

1) Problem Setup: We first consider a small scale portfolio optimization problem with totally five stocks and we aim to find the three particular stocks to form an optimal portfolio with the minimum value for the combined cost taking both risk and return into consideration. In this illustrative experiment, the mean return r_0 and the variance matrix Σ are given as

$$r_0 = \begin{bmatrix} 0.0007 & 0.0004 & 0.0013 & 0.0014 & 0.0017 \end{bmatrix}^{\mathrm{T}}$$

$$\Sigma = 10^{-3} \times \begin{bmatrix} 0.0970 & 0.0361 & 0.0376 & 0.0283 & 0.0341 \\ 0.0361 & 0.0619 & 0.0257 & 0.0230 & 0.0229 \\ 0.0376 & 0.0257 & 0.1264 & 0.0321 & 0.0254 \\ 0.0283 & 0.0230 & 0.0321 & 0.1413 & 0.0436 \\ 0.0341 & 0.0229 & 0.0254 & 0.0436 & 0.1138 \end{bmatrix}$$

based on stock data of five DJI companies, namely, American express company (AXP), general electric (GE), McDonald's Co. (McD), Merck & Co. Inc. (MRK), and AT&T Inc. (AT&T), for the year 2006. The transaction charge rate is set at 0.01. The risk and return tradeoff factor c_0 is set as 0.1.

2) Neural Solution: As to the neural network parameters, the scaling factor ϵ is set as 10^{-7}, the boolean constraint enforcement factor c_1 is set as 10^{-4}, the equality enforcement factor c_2 is set as 5×10^{-3}, the sigmoid function coefficient c_3 is set as 0.01. As shown in Fig. 2, the neural variables x and z converge with time. The steady state values of are $x = [9.6 \times 10^{-26}, 3.5 \times 10^{-28}, 0.27, 0.18, 0.54]$ and $z = [3.4 \times 10^{20}, 2.1 \times 10^{-20}, 1.0, 1.0, 1.0]$, which implies the selection of stocks 3 (McD), 4 (MRK), and 5 (AT&T) for optimal investment. There are totally ten combinations

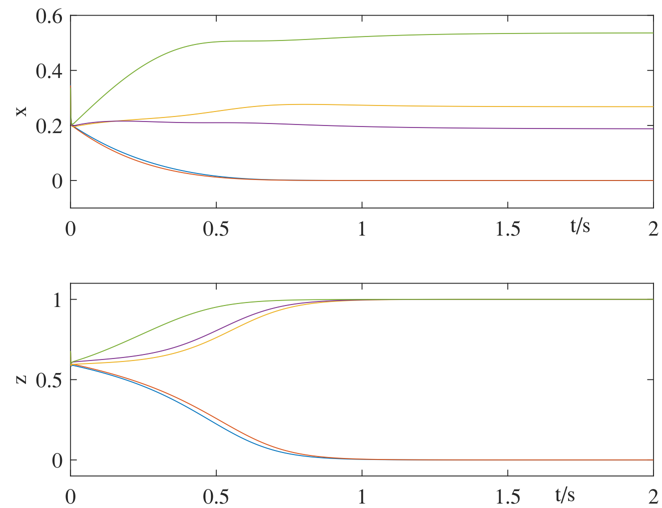

Fig. 2. Transient behavior of neural variables x and z of the neural variables in the small scale illustrative experiment.

for the selection of three stocks from five in total. For each determined combination, the portfolio optimization degrades to standard Markowitz formulation in the form of constrained quadratic program with both equality and inequality constraints being linear and therefore can be readily solved using convex optimization. After enumerating the ten combinations, the global optimal solution can be obtained as the selection of stock three, four, and five with weights 0.27, 0.19, and 0.54, which is consistent with the solution obtained by the proposed method with a reasonable error torlerance. For each of the ten possible selections, the optimal weights can be calculated and the minimum cost can therefore obtained. The average minimum cost of the ten selections is calculated as -6.3×10^{-5} while the optimum cost obtained by the proposed method is -8.7×10^{-5}, which improves by 38.1% over the average. In contrast to the convex solution based on enumeration method, the proposed method directly outputs one optimal solution without any enumeration, which significantly reduces computational time consumption.

3) Impact of Parameters: In this part, we evaluate the impact of different parameters to the behavior of the system. Note that the risk-return tradeoff parameters c_0, the transaction cost rate t_0, the number of selected stocks k_0, the mean return r_0 and the variance matrix Σ are determined by specific applications and will be set by practitioners for their values. The neural coefficients ϵ, c_1, c_2, and c_3 are to be set as system parameters. Fig. 3 shows the comparisons of the behaviors of x and z with different values for $\epsilon = 10^{-5}$ and $\epsilon = 10^{-6}$ in contrast to $\epsilon = 10^{-7}$ in Fig. 2 without changing the values for other parameters. It is observable that ϵ determines the time scale of the evolution of the system and a smaller value of ϵ reaches a faster evolution for the neural network. c_1 is a penalty coefficient to enforce the boolean constraint of variable z. As this term appears in the objective function as a penalty, we expect its influence to the optimality of the original objective function before introducing this term is minimal. Therefore, a small value of c_1 is preferred. Fig. 4 shows the dynamic behaviors of x and z when choosing different values

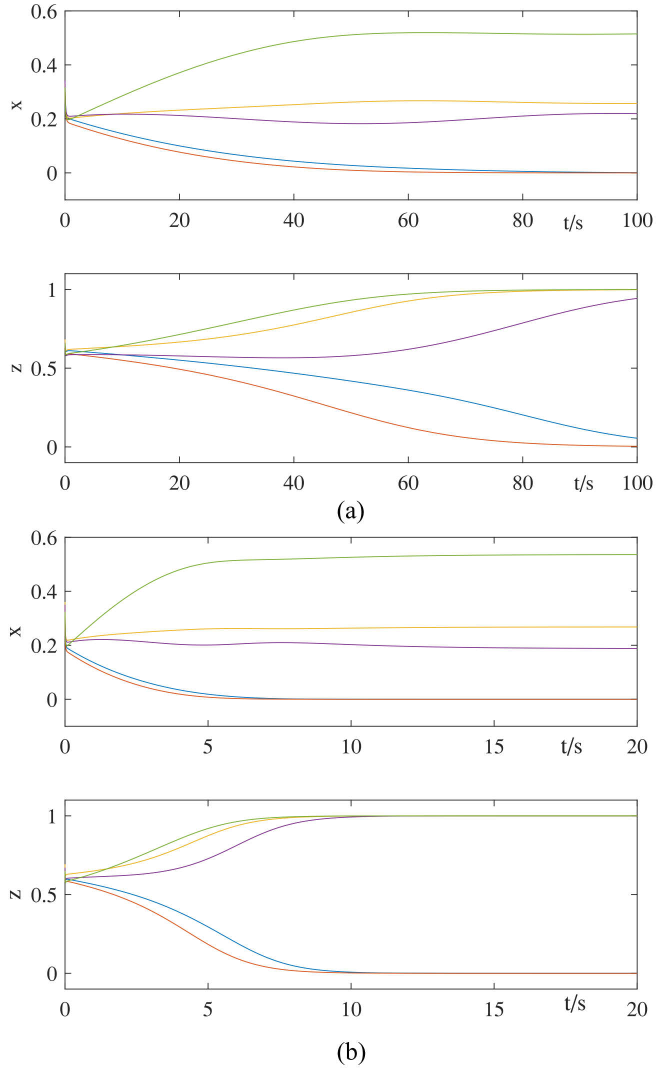

Fig. 3. Impact of neural parameter ϵ to the transient behavior of the neural variables x and z with (a) $\epsilon = 10^{-5}$, (b) $\epsilon = 10^{-6}$ in contrast to $\epsilon = 10^{-7}$ for the case in Fig. 2.

of c_1, i.e., $c_1 = 10^{-2}$, $c_1 = 10^{-3}$, and $c_1 = 10^{-5}$ (the result associated with $c_1 = 10^{-4}$ is shown in Fig. 2. A clear observation is that the increase of c_1 speeds up the convergence and reduces the transient time. However, a close look of the curves can find that the steady state values of x in Fig. 4(a) differs greatly from others. Actually, the case with $c_1 = 10^{-2}$ converges to nonoptimal solution due to the influence of the penalty term. Of course, too small values of c_1 leads to the ignorable enforcement of the boolean constraint of z, making z converge to nonboolean values. Therefore, an appropriate value of c_1 is necessary and it is found that $c_1 = 10^{-4}$ is a good choice. The parameter c_2 brings an equality constraint into a penalty term in the objective function. Similar to the scenario for c_1, an appropriate value of c_2 is preferred, not too big to impact the original objective function too much, not too ignorably small to lose the enforcement of the equality constraint. Actually, the system is less sensitive to the value of c_2 than to c_1. For a large range of c_2, as tested in numerical experiments, the setting of c_2 ranging from $c_2 = 1.0 \times 10^{-4}$ to $c_2 = 100$, all returns satisfactory results. The impact of c_3 to system transient is shown in Fig. 5 with $c_3 = 10^{-1}$, 10^{-3} and 10^{-4} in contrast to $c_3 = 10^{-2}$ for the case in Fig. 2. It is observable that the increase of c_3 speeds up the converge while the steady state value of x in Fig. 5(a) with the largest c_3 differs from the optimal solution found by the enumeration. Choosing a smaller value of c_3 help avoid falling into local optima while greater value of c_3 expedite the convergence.

B. Portfolio Selection With All DJI Stocks

In this part, we apply the proposed method to the optimal portfolio analysis with the selection of totally five stocks from all DJI stocks. Without losing generality, we consider the DJI stock data in the year 2006 and use the same parameter setup as in the illustrative experiment.

Numerical experiments are conducted and Fig. 6 shows the dynamics of neural variable x and z of typical experimental run. From this figure, it is observable that the values converges to the steady state within 5 s with the output of steady state values for x and z being $x = [1.0249 \times 10^{-43}, 3.9643 \times 10^{-38}, 5.205 \times 10^{-32}, 1.9656 \times 10^{-23}, 6.0965 \times 10^{-27}, 1.1544 \times 10^{-38}, 8.7838 \times 10^{-27}, 0.21062, 3.5577 \times 10^{-32}, 0.077518, 1.9262 \times 10^{-46}, 5.1158 \times 10^{-28}, 5.8236 \times 10^{-14}, 1.2177 \times 10^{-26}, 8.9646 \times 10^{-68}, 1.3344 \times 10^{-27}, 9.4204 \times 10^{-29}, 3.3779 \times 10^{-24}, 1.785 \times 10^{-14}, 6.7375 \times 10^{-37}, 7.5904 \times 10^{-22}, 0.12463, 2.2605 \times 10^{-30}, 2.8912 \times 10^{-33}, 1.3508 \times 10^{-29}, 0.39131, 9.2491 \times 10^{-33}, 1.7924 \times 10^{-23}, 3.9366 \times 10^{-43}, 0.19055]$, and $z = [2.7905 \times 10^{-19}, 3.0735 \times 10^{-19}, 3.5306 \times 10^{-19}, 4.4043 \times 10^{-19}, 4.049 \times 10^{-19}, 2.8089 \times 10^{-19}, 4.0845 \times 10^{-19}, 1, 3.3649 \times 10^{-19}, 1, 2.8443 \times 10^{-19}, 3.7388 \times 10^{-19}, 5.824 \times 10^{-14}, 4.4357 \times 10^{-19}, 2.592 \times 10^{-19}, 4.1151 \times 10^{-19}, 3.6527 \times 10^{-19}, 5.8664 \times 10^{-19}, 1.7895 \times 10^{-14}, 2.9211 \times 10^{-19}, 7.8919 \times 10^{-19}, 1, 3.8174 \times 10^{-19}, 3.2922 \times 10^{-19}, 3.7958 \times 10^{-19}, 1, 3.4781 \times 10^{-19}, 7.8496 \times 10^{-19}, 2.9877 \times 10^{-19}, 1]$, with the compliance to the equality constraint $\mathbf{1}^{\mathrm{T}}x + {t_0}^{\mathrm{T}}x = 1$ with an error for 4.5736×10^{-3}, the equality constraint $\mathbf{1}^{\mathrm{T}}z = k_0$ with an error for $\mathbf{1}^{\mathrm{T}}z - k_0 = 7.6383 \times 10^{-14}$, and strict compliance to the inequality constraint $0 \le x \le z$ and boolean values for the steady state of z with the maximum error 5.824×10^{-14}. The solution indicates the selection of the 8th, the 10th, the 22th, the 26th, and the 30th stocks with weights, respectively $[0.21062, 0.077518, 0.12463, 0.39131, 0.19055]$. By running the proposed algorithm with ten trials with random initialization, the variables x and z converges to identical values within a tolerance range of the error indicating the selection of the same stocks with the same portfolio across different runs. To compare with the baseline, we enumerate all possible combinations by choosing five stocks out of all 30 stocks in DJI with a total number of $\binom{30}{5} = 1.42506 \times 10^5$ combinations and with the resolution of each combination by calling MATLAB routine quadprog() to solve 1.42506×10^5 individual constrained quadratic optimizations. The program is run on a computer with Intel Core i7-8550U CPU at frequencies 1.80 GHz and 1.99 GHz, and RAM for 16.0 GB and it takes 944.208709 s to complete the enumeration

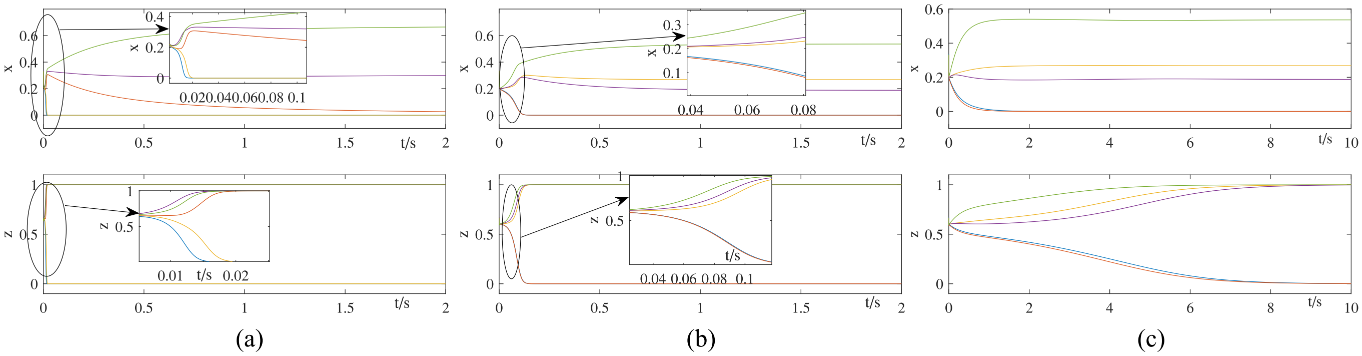

Fig. 4. Impact of neural parameter c_1 to the transient behavior of the neural variables x and z with (a) $c_1 = 10^{-2}$, (b) $c_1 = 10^{-3}$, (c) $c_1 = 10^{-5}$ in contrast to $c_1 = 10^{-4}$ for the case in Fig. 2.

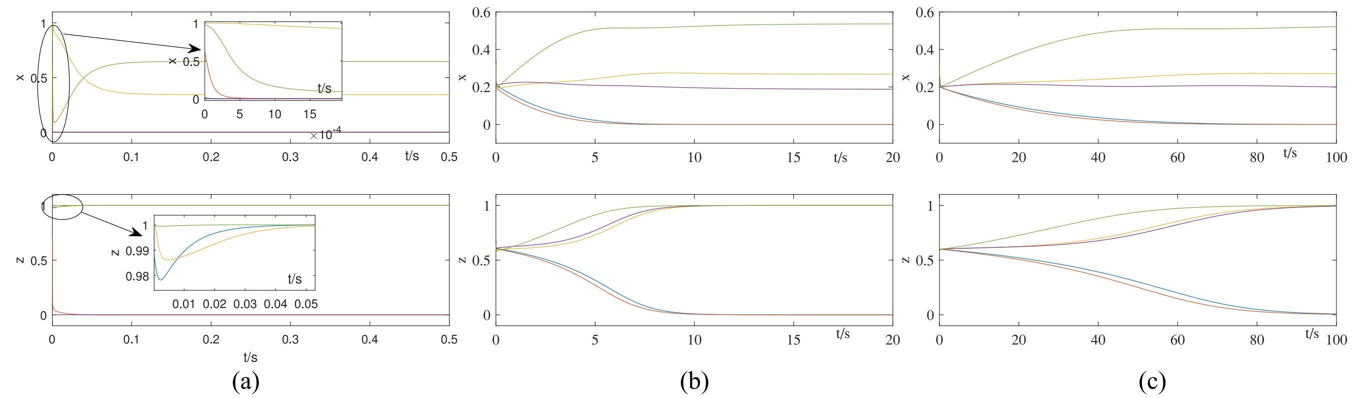

Fig. 5. Impact of neural parameter c_3 to the transient behavior of the neural variables x and z with (a) $c_3 = 10^{-1}$, (b) $c_3 = 10^{-3}$, (c) $c_3 = 10^{-4}$ in contrast to $c_3 = 10^{-2}$ for the case in Fig. 2.

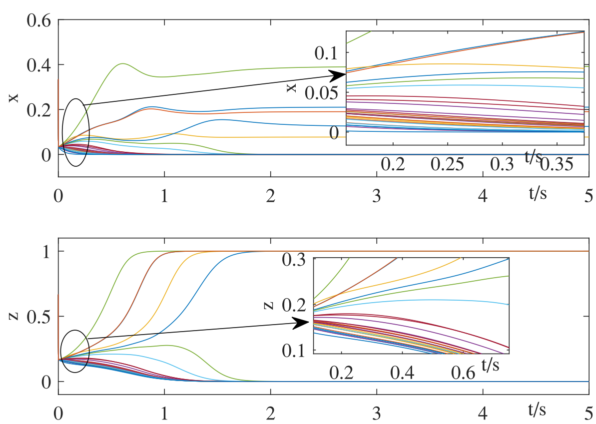

Fig. 6. Transient behavior of neural variables x and z of the neural variables for the portfolio optimization with Cardinality constraints with all DJI stocks in the year of 2006.

with the return of the globally optimal solution by choosing stocks 8, 10, 22, 26, and 30 with weights 0.20947, 0.077823, 0.12374, 0.38983, and 0.18924 with the total combined risk-return cost -1.0173×10^{-4}. Clearly, the solution obtained by the proposed method is highly consistent with the global optimum but significantly reduces the computational time consumption. The average cost can be obtained by averaging the cost of each individual objective function over the $\binom{30}{5}$ combinations of stocks and the value is -4.549×10^{-5}. Clearly, the optimal portfolio obtained using the proposed method with the cost -1.0173×10^{-4} is reduced by 123.6% in terms of the combined risk-return cost, which demonstrates the effectiveness of the proposed strategy.

VII. Conclusion and Future Work

In this article, we proposed a novel neural network solution to the portfolio optimization problem under both transaction cost and cardinality constraints. The nonconvex cardinality constraint poses great challenges and this work leverages the recent progress on dynamic neural networks to tackle the problem. Rigorous proof is provided to show the global convergence of the proposed model. Numerical experiments demonstrates the effectiveness of the proposed approach. The portfolio optimization with cardinality constraint formulated in this article is a nonconvex optimization problem and usually includes more than one local optimum. The proposed dynamic neural network is able to find one local optimum solution of the optimization and cannot guarantee the found solution to be a global optimum, although numerical experiments show that the obtained solution outperforms the one by the existing MATLAB routine. This constructs one limitation of this article. Further research on this topic is the design of new neural dynamics to identify the global optimum. Another possible future research direction is the extension of this work to general constrained nonconvex optimization problems and the application of it to solve more practical problems. Before ending this article, it worths pointing out that this is the first dynamic neural model for portfolio optimization with the consideration of both transaction cost and cardinality constraints and the proposed model reaches 123.6% less cost in comparison to the averagely selected portfolios as shown in our experiment based on real stock data in the year of 2006 for all DJI listed companies.

REFERENCES

[1] H. M. Markowitz, "Foundations of portfolio theory," *J. Financ.*, vol. 46, no. 2, pp. 469–477, 1991.

[2] I. Omisore, M. Yusuf, and N. Christopher, "The modern portfolio theory as an investment decision tool," *J. Account. Taxation*, vol. 4, no. 2, pp. 19–28, 2011.

[3] D. Kim and J. C. Francis, *Modern Portfolio Theory: Foundations, Analysis, and New Developments*. Hoboken, NJ, USA: Wiley, 2013.

[4] A. T. Khan, X. Cao, S. Li, V. N. Katsikis, I. Brajevic, and P. S. Stanimirovic, "Fraud detection in publicly traded U.S firms using beetle antennae search: A machine learning approach," *Expert Syst. Appl.*, vol. 191, pp. 116–148, Apr. 2022.

[5] A. H. Khan, X. Cao, and S. Li, *Management and Intelligent Decision-Making in Complex Systems: An Optimization-Driven Approach*. Singapore: Springer Nat., 2020.

[6] J. A. Adamu, "Advanced stochastic optimization algorithm for deep learning artificial neural networks in banking and finance industries," *Risk Financ. Manage.*, vol. 1, no. 1, pp. 8–28, 2019.

[7] L. Troiano, E. M. Villa, and V. Loia, "Replicating a trading strategy by means of LSTM for financial industry applications," *IEEE Trans. Ind. Informat.*, vol. 14, no. 7, pp. 3226–3234, Jul. 2018.

[8] A. T. Khan, X. Cao, Z. Li, and S. Li, "Enhanced beetle antennae search with zeroing neural network for online solution of constrained optimization," *Neurocomputing*, vol. 447, pp. 294–306, Aug. 2021.

[9] W. Gao and C. Su, "Analysis on block chain financial transaction under artificial neural network of deep learning," *J. Comput. Appl. Math.*, vol. 380, pp. 112–131, Dec. 2020.

[10] I. Goldstein, W. Jiang, and G. A. Karolyi, "To FinTech and beyond," *Rev. Financ. Stud.*, vol. 32, no. 5, pp. 1647–1661, 2019.

[11] F. Allen, X. Gu, and J. Jagtiani, "A survey of FinTech research and policy discussion," *Rev. Corporate Financ.*, vol. 1, no. 20–21, pp. 259–339, 2021.

[12] P. Rebentrost, B. Gupt, and T. R. Bromley, "Quantum computational finance: Monte carlo pricing of financial derivatives," *Phys. Rev. A*, vol. 98, no. 2, pp. 302–321, 2018.

[13] D. T. Tran, A. Iosifidis, J. Kanniainen, and M. Gabbouj, "Temporal attention-augmented bilinear network for financial time-series data analysis," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 30, no. 5, pp. 1407–1418, May 2019.

[14] A. Tsantekidis, N. Passalis, A.-S. Toufa, K. Saitas-Zarkias, S. Chairistanidis, and A. Tefas, "Price trailing for financial trading using deep reinforcement learning," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 32, no. 7, pp. 2837–2846, Jul. 2021.

[15] S. Sharma, N. Islam, G. Singh, and A. Dhir, "Why do retail customers adopt artificial intelligence (AI) based autonomous decision-making systems," *IEEE Trans. Eng. Manag.*, early access, Apr. 18, 2022, doi: 10.1109/TEM.2022.3157976.

[16] M. H. Davis and A. R. Norman, "Portfolio selection with transaction costs," *Math. Oper. Res.*, vol. 15, no. 4, pp. 676–713, 1990.

[17] W. Zhang, Y. Liu, and W. Xu, "A possibilistic mean-semivariance-entropy model for multi-period portfolio selection with transaction costs," *Eur. J. Oper. Res.*, vol. 222, no. 2, pp. 341–349, 2012.

[18] K. Imajo, K. Minami, K. Ito, and K. Nakagawa, "Deep portfolio optimization via distributional prediction of residual factors," in *Proc. AAAI Conf. Artif. Intell.*, 2021, pp. 213–222.

[19] T.-J. Chang, N. Meade, J. E. Beasley, and Y. M. Sharaiha, "Heuristics for cardinality constrained portfolio optimisation," *Comput. Oper. Res.*, vol. 27, no. 13, pp. 1271–1302, 2000.

[20] J. Gao, D. Li, X. Cui, and S. Wang, "Time cardinality constrained mean–variance dynamic portfolio selection and market timing: A stochastic control approach," *Automatica*, vol. 54, pp. 91–99, Apr. 2015.

[21] P. J. Kremer, S. Lee, M. Bogdan, and S. Paterlini, "Sparse portfolio selection via the sorted ℓ1-norm," *J. Bank. Financ.*, vol. 110, pp. 105–137, Jan. 2020.

[22] R. Raei and H. Alibeiki, "Portfolio optimization using particle swarm optimization method," *Financ. Res. J.*, vol. 12, no. 29, pp. 1–20, 2010.

[23] H. Zhu, Y. Wang, K. Wang, and Y. Chen, "Particle swarm optimization (PSO) for the constrained portfolio optimization problem," *Expert Syst. Appl.*, vol. 38, no. 8, pp. 10161–10169, 2011.

[24] M. Kaucic, "Equity portfolio management with cardinality constraints and risk parity control using multi-objective particle swarm optimization," *Comput. Oper. Res.*, vol. 109, pp. 300–316, Sep. 2019.

[25] I. Yaman and T. E. Dalkiliç, "A hybrid approach to cardinality constraint portfolio selection problem based on nonlinear neural network and genetic algorithm," *Expert Syst. Appl.*, vol. 169, pp. 401–517, May 2021.

[26] C.-H. Chen, Y.-H. Chen, J. C.-W. Lin, and M.-E. Wu, "An effective approach for obtaining a group trading strategy portfolio using grouping genetic algorithm," *IEEE Access*, vol. 7, pp. 7313–7325, 2019.

[27] A. T. Khan, X. Cao, S. Li, B. Hu, and V. N. Katsikis, "Quantum beetle antennae search: A novel technique for the constrained portfolio optimization problem," *Sci. China Inf. Sci.*, vol. 64, no. 5, pp. 1–14, 2021.

[28] V. N. Katsikis, S. D. Mourtas, P. S. Stanimirović, S. Li, and X. Cao, "Time-varying minimum-cost portfolio insurance under transaction costs problem via beetle antennae search algorithm (BAS)," *Appl. Math. Comput.*, vol. 385, Nov. 2020, Art. no. 125453.

[29] A. H. Khan et al., "Optimal portfolio management for engineering problems using nonconvex cardinality constraint: A computing perspective," *IEEE Access*, vol. 8, pp. 57437–57450, 2020.

[30] M. A. Medvedeva, V. N. Katsikis, S. D. Mourtas, and T. E. Simos, "Randomized time-varying knapsack problems via binary beetle antennae search algorithm: Emphasis on applications in portfolio insurance," *Math. Methods Appl. Sci.*, vol. 44, no. 2, pp. 202–212, 2021.

[31] Y. Zhang and S. S. Ge, "Design and analysis of a general recurrent neural network model for time-varying matrix inversion," *IEEE Trans. Neural Netw.*, vol. 16, no. 6, pp. 1477–1490, Nov. 2005.

[32] D. Guo and Y. Zhang, "Novel recurrent neural network for time-varying problems solving," *IEEE Comput. Intell. Mag.*, vol. 7, no. 4, pp. 61–65, Nov. 2012.

[33] Y. Shi and Y. Zhang, "New discrete-time models of zeroing neural network solving systems of time-variant linear and nonlinear inequalities," *IEEE Trans. Syst., Man, Cybern., Syst.*, vol. 50, no. 2, pp. 565–576, Feb. 2020.

[34] Y. Zhang, M. Yang, H. Huang, J. Chen, and Z. Li, "Unified solution of different-kind future matrix equations using new nine-instant discretization formula and zeroing neural dynamics," *IEEE Trans. Syst., Man, Cybern., Syst.*, vol. 52, no. 8, pp. 4993–5003, Aug. 2022.

[35] Z. Ma and D. Guo, "Discrete-time recurrent neural network for solving bound-constrained time-varying underdetermined linear system," *IEEE Trans. Ind. Informat.*, vol. 17, no. 6, pp. 3869–3878, Jun. 2021.

[36] J. Li, Y. Zhang, and M. Mao, "Continuous and discrete zeroing neural network for different-level dynamic linear system with robot manipulator control," *IEEE Trans. Syst., Man, Cybern., Syst.*, vol. 50, no. 11, pp. 4633–4642, Nov. 2020.

[37] B. Liao, Y. Zhang, and L. Jin, "Taylor $O(h^3)$ discretization of ZNN models for dynamic equality-constrained quadratic programming with application to manipulators," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 27, no. 2, pp. 225–237, Feb. 2016.

[38] Y. Zhang, S. Li, J. Weng, and B. Liao, "GNN model for time-varying matrix inversion with robust finite-time convergence," *IEEE Trans. Neural Netw. Learn. Syst.*, early access, May 24, 2022, doi: 10.1109/TNNLS.2022.3175899.

[39] D. Chen, S. Li, F.-J. Lin, and Q. Wu, "New super-twisting zeroing neural-dynamics model for tracking control of parallel robots: A finite-time and robust solution," *IEEE Trans. Cybern.*, vol. 50, no. 6, pp. 2651–2660, Jun. 2020.

[40] Z. Zhang et al., "Robustness analysis of a power-type varying-parameter recurrent neural network for solving time-varying QM and QP problems and applications," *IEEE Trans. Syst., Man, Cybern., Syst.*, vol. 50, no. 12, pp. 5106–5118, Dec. 2020.

[41] L. Xiao, J. Tao, J. Dai, Y. Wang, L. Jia, and Y. He, "A parameter-changing and complex-valued zeroing neural-network for finding solution of time-varying complex linear matrix equations in finite time," *IEEE Trans. Ind. Informat.*, vol. 17, no. 10, pp. 6634–6643, Oct. 2021.

[42] Z. Li, B. Liao, F. Xu, and D. Guo, "A new repetitive motion planning scheme with noise suppression capability for redundant robot manipulators," *IEEE Trans. Syst., Man, Cybern., Syst.*, vol. 50, no. 12, pp. 5244–5254, Dec. 2020.

[43] L. Xiao, Y. He, and B. Liao, "A parameter-changing zeroing neural network for solving linear equations with superior fixed-time convergence," *Expert Syst. Appl.*, vol. 208, Dec. 2022, Art. no. 118086.

[44] Y. Zhang, S. Li, and X. Liu, "Neural network-based model-free adaptive near-optimal tracking control for a class of nonlinear systems," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 29, no. 12, pp. 6227–6241, Dec. 2018.

[45] Z. Zhang, L. Zheng, H. Yang, and X. Qu, "Design and analysis of a novel integral recurrent neural network for solving time-varying Sylvester equation," *IEEE Trans. Cybern.*, vol. 51, no. 8, pp. 4312–4326, Aug. 2021.

[46] Y. Qi, L. Jin, X. Luo, and M. Zhou, "Recurrent neural dynamics models for perturbed nonstationary quadratic programs: A control-theoretical perspective," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 33, no. 3, pp. 1216–1227, Mar. 2022.

[47] Y. Shi, Z. Pan, J. Li, B. Li, and X. Sun, "Recurrent neural dynamics for handling linear equation system with rank-deficient coefficient and disturbance existence," *J. Franklin Inst.*, vol. 359, no. 7, pp. 3090–3102, 2022.

[48] S. Li, Y. Zhang, and L. Jin, "Kinematic control of redundant manipulators using neural networks," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 28, no. 10, pp. 2243–2254, Oct. 2017.

[49] L. Liu, X. Li, Y.-J. Liu, and S. Tong, "Neural network based adaptive event trigger control for a class of electromagnetic suspension systems," *Control Eng. Pract.*, vol. 106, Jan. 2021, Art. no. 104675.

[50] D. Chen, S. Li, and L. Liao, "A recurrent neural network applied to optimal motion control of mobile robots with physical constraints," *Appl. Soft Comput.*, vol. 85, pp. 105–120, Dec. 2019.

[51] L. Liu, W. Zhao, Y.-J. Liu, S. Tong, and Y.-Y. Wang, "Adaptive finite-time neural network control of nonlinear systems with multiple objective constraints and application to electromechanical system," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 32, no. 12, pp. 5416–5426, Dec. 2021.

[52] S. Li, J. He, Y. Li, and M. U. Rafique, "Distributed recurrent neural networks for cooperative control of manipulators: A game-theoretic perspective," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 28, no. 2, pp. 415–426, Feb. 2017.

[53] L. Jin, S. Li, B. Liao, and Z. Zhang, "Zeroing neural networks: A survey," *Neurocomputing*, vol. 267, pp. 597–604, Dec. 2017.

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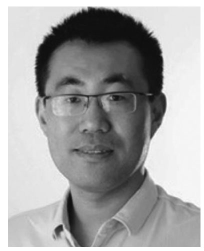

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